

Dynamic latent-state and performance-regime explorer

Can a better current horse state be inferred by treating historical performances as noisy, context-dependent measurements rather than equally trustworthy observations?

Parts II and IV of the HKJC latent-performance programme

1 September 2026

What this document is. The report the plan HKJC_DYNAMIC_LATENT_STATE_REGIME_EXPLORER.md asks for, answering all forty-two of its sections. It follows that document's own §39 structure.

Why there are two reports for one brief. The plan document was not on disk when this work began; Part II was executed from a nine-theme summary and delivered as `report/dynamic-state.pdf`. The document was supplied on 1 September 2026, and reading it against the delivered work identified **six substantive sections that had never been executed** — latent switching regimes, the multidimensional sectional state, jockey interactions, metric-learned peer pools, per-run predictive information, and the residual mixture as a measurement model. Part IV executed them. This report covers the whole brief; `dynamic-state.pdf` remains as the Part II artifact.

What it does not establish. No profitability claim. Every price is a result-page price recorded after the pool closed. No dividends are loaded and no return is computed anywhere. Every season has been development data.

The one-line answer. Historical races *are* noisy, context-dependent measurements, and the noise *is* partly identifiable — a four-component mixture on the state residual separates runs whose trouble rate runs 0.06 to 0.23 and whose information content runs +0.075 to **-0.078**, and using it as the measurement variance improves the latent-state proxy by +0.008 against a control of +0.003. **And none of it survives canonical scale:** that block returns +0.0017 in production, worse than adding nothing. Two of the failures are informative — the latent regime is **unidentifiable** at this signal-to-noise, and variance-maximising factor extraction on sectionals recovers **style rather than ability** by construction — and one is a correction: the first canonical run of this part reported -0.475, which was **leakage**, found and fixed by noticing the number was 105 times the production edge.

Contents

1	Executive summary	3
2	What the previous study already settled	4
3	Data and forward-only construction	5
4	Observation and run regimes	5
5	Sectional evidence of run quality	6
6	Dynamic scalar horse state	6
7	Variable observation reliability	7
7.1	Per-run predictive information (§29)	7
7.2	The residual mixture as a measurement model (§30)	7

8	Pace-adjusted historical performance	8
9	Jockey and environment corrections	8
9.1	Jockey (§16)	8
9.2	Environment (§17)	9
10	Opponent adjustment	10
11	Multidimensional sectional state (§13)	10
12	State-transition regimes (§12, §20)	11
12.1	Is the latent regime identifiable here at all?	11
12.2	The switching state on real data	12
12.3	Process archetypes (§20)	12
13	Dynamic analog and peer pooling (§22, §23)	12
14	Filter diagnostics, and a defect four parts of work had missed	14
14.1	What passes	14
14.2	§37.9 fails: the transition is misspecified	14
14.3	The repair works, and forecasts no better	15
15	Sparse-history horses (§26)	15
16	Market and handicap disagreement (§19)	17
17	Production integration	17
18	Canonical confirmation	18
19	Limitations	19
20	Recommendations	20

1 Executive summary

The brief's §39 asks ten specific questions. Answering them in order, with the evidence behind each named:

1. **Is a historical race a noisy measurement of latent ability?** **Yes, and demonstrably so.** A four-component mixture on the state residual (D10) separates components whose severe-trouble rates run 0.061, 0.078, 0.124, 0.232 and whose per-run information content runs +0.075, +0.047, +0.021, −0.078 — monotonically aligned, and discovered without supervision. **44% of a horse's own past runs make the prediction of its next run worse.**
2. **Can run quality / information content be predicted?** **Yes, and barely.** The per-run leave-one-out information I_{ir} is predictable at out-of-sample $r = 0.099$ against a permuted control of 0.0006 — real, and 0.7% of its variance. No single driver reaches $|\rho| = 0.04$.
3. **Does downweighting compromised runs improve current-state estimation?** **On the latent-state proxy, yes** — +0.00803 of forward correlation against a permuted control of +0.00285, when the downweighting comes from the mixture placed in the measurement model. **In production, no:** §18 puts that block at +0.001718, $t = +1.46$, on 15 of 40 cells. This is the brief's §30, and it is the part's clearest instance of a real mechanism with negative production value. Note also the tension with Part III's oracle result, resolved in §7.3.
4. **Does dynamic state beat simple recency weighting?** **Yes** — Part II's ladder: static prior mean 0.277, 60-day EWM 0.351, heteroskedastic Kalman 0.354.
5. **Are there distinct state-transition regimes?** **Not identifiably.** $K = 1$ beats $K = 2, 3, 4$ on forward correlation; regime shares stay within a per cent of uniform; the regime posterior sits at **1.095 nats against a maximum of $\log 3 = 1.0986$** and does not fall between a horse's debut and its thirtieth start. A simulation ladder shows why: at HKJC's observation noise the regime posterior tracks a *known* simulated regime at $r = 0.028$.
6. **Does local peer borrowing help sparse horses?** **Locality is unambiguously real** — every neighbourhood construction beats a random size-matched cohort by an order of magnitude ($r(\text{peer mean } \Delta A) + 0.35$ against +0.01). Metric learning helps at small pool sizes and stops mattering by $k = 200$.
7. **Does opponent adjustment help?** **No** — Part II: −0.058, worse than its own permutation control.
8. **Do full sectionals / 200 m splits improve state inference?** **Marginally, and not as a state.** 200 m resolution adds $\approx +0.002$ over the 400 m bands (replicated twice independently). The sectional *state vector* §13 asks for is worth +0.0005; the same factors *unfiltered* are worth +0.055.
9. **Does jockey/context correction improve historical state?** **No.** Support is more than adequate (98.5% of rides in cells with ≥ 30), the effects are real (jockey carries 3.74% of target variance), and the **permuted-jockey control beats every real correction.**
10. **Which final features improve production?** See §18.

And a defect the required-figure list caught

§37's list of "required figures" earned its place. Drawing innovations against the time gap — a diagnostic no earlier stage had produced — shows the mean innovation running −0.031 at a fortnight to **−0.455** beyond a year, monotone across a fifteen-fold range. **Part II's transition is**

misspecified: it reverts the state toward a global mean, and horses returning from a layoff run well below their own decayed state.

A gap-dependent offset *fixes* it — correlation from -0.098 to $+0.002$ — and forecasts no better: $+0.00148$ against its own shuffled control's $+0.00456$. That is the fourth confound in this programme that is real, correctly removed, and still lost to permutation. **Correct specification and forecasting skill are different objectives.**

The two findings worth carrying forward

Persistence and usefulness are inverted across the sectional profile. The earliest 400 m section is the *most* self-persistent quantity measured ($+0.409$) and the *least* predictive of the next performance ($+0.020$) — a ratio of 0.048. The final section is the reverse ($+0.272$ and $+0.240$, ratio 0.880). A horse's early habits are the most reliably measured thing about it and nearly worthless for forecasting.

Variance-maximising factor extraction on sectionals is counterproductive, not neutral. The first principal direction holds **75.7%** of the sectional profile's variance and that direction is style. So factor analysis preferentially recovers the one component §3.7 warns against mistaking for ability, and the extracted factors predict the next performance **negatively** (-0.131 , -0.197) while predicting their own futures at $+0.32$. This is a structural argument against a whole class of approach.

2 What the previous study already settled

The brief's §3 lists nine things as prior evidence rather than open questions, and its §42 insists this must not be another grouping study. Both constraints were respected: no conventional static horse grouping was rerun in Part II or Part IV.

Carried forward from Parts I–III as settled, and used rather than retested:

- Sections are exactly comparable across distances — 400 m back from the finish, first carrying the remainder, valid on 1.0000 of 13,309 races.
- The target is y_z^{rel} , the within-race z -score of log average speed: self-persistence 0.278, split-half reliability 0.822.
- Horse ability drifts with a ≈ 60 -day half-life; a static career estimate is the worst of eight tested.
- Jockey identity forecasts relative performance about four times better than horse identity (oos R^2 0.048 against 0.012).
- Novelty and usefulness are inversely related (Spearman $+0.39$), reconfirmed three times since.
- An oracle reliability built from realised surprise is worth $+0.106$ of forward correlation; observables reach 0.27% of it.
- **Fourteen candidates have been carried to canonical scale across the programme and none has confirmed at $t < -2$.**

Which sections of this brief were and were not executed before Part IV

Section	What it asks	What existed before Part IV
§12, §20, Model 8	latent switching regimes, estimated Π and K	nothing — d02 was handed <i>observable calendar events</i> whose only effect was a multiplier on q
§13, Model 7	a sectional state <i>vector</i> by factor analysis, filtered	nothing — d02 filters one scalar
§16	jockey $\times$ regime and jockey $\times$ style	nothing — zero occurrences of "jockey" in d03
§22 (full), §23	five neighbourhood constructions compared; a <i>learned</i> metric	partial — one bucketed score on a 1-D locality index
§29	per-run I_{ir} , and whether it is <i>predictable</i>	partial — an informativeness penalty by four-way tier
§30, §31	the residual mixture in the <i>measurement</i> model; three uncertainty channels	nothing — Part I fitted mixtures to <i>winner shocks</i> , the opposite placement

3 Data and forward-only construction

123,973 runner-rows, 2008–2026, Class 1–5. Development before 2020, evaluation 2020–2026. Canonical confirmation on the reference project’s own eight-year, five-seed, 400-round profile.

Every history-derived quantity is **date-atomic**: the forward pass reads each horse’s state before applying any of the current *date*’s observations, not merely the current row’s. That distinction is load-bearing — a per-row shift on a date-sorted frame excludes the previous *row*, which on a race day is another runner in the same race. Every fitted coefficient is fitted on years strictly earlier than the year it scores.

The switching filter of §5 inherits the same discipline: state, variance and regime posterior are all emitted before absorption, so two runners on one card cannot see each other and no horse sees its own same-day result.

One correction to the record

The first version of the switching module’s EM produced a plausible null for the wrong reason. Its M-step regressed each regime’s "current" state on its "previous" state — but the Gaussian-sum collapse leaves every regime sharing one posterior mean per row, so that regression is identical across regimes by construction and identifies nothing. The regimes collapsed onto a uniform fixed point: transition rows equal to three decimals, both innovation variances at 1.94, and a regime posterior tracking simulated truth at $r = 0.017$. The fit now maximises the one-step predictive log-likelihood by coordinate ascent, and §5’s identifiability ladder is the evidence that the corrected version works. The null in §12 is a real null, not that bug.

4 Observation and run regimes

Part II built the observation model: stewards’ comments as weak labels (severe trouble, moderate trouble, none recorded), pace context, sectional anomaly, and a per-run observation variance fitted by regressing the *squared forward surprise* on run characteristics — deliberately not on the run’s own residual, which is contaminated by the state it is trying to measure.

Its discipline governs Part IV too: a correction earns its place only if the corrected run predicts the same horse’s *next* run better than the uncorrected one, and every correction is tested against its own within-year permutation.

Part III added the extreme-run studies the brief’s §27 and §28 ask for. Retention at 2 sd:

Direction	Reason	n	This run	Retention
bad	recorded trouble	1,506	−2.599	+0.288
bad	sectional anomaly	3,260	−2.607	+0.281
bad	pace mismatch	1,142	−2.486	+0.383
bad	none identified	2,097	−2.356	+0.377
good	none identified	371	+2.178	+0.198

Two asymmetries. **Bad runs are more informative than good runs** (0.28–0.38 against 0.20–0.23). And explained bad runs are slightly *less* informative than unexplained ones, which is §27’s hypothesis confirmed in direction and far too small to pay for downweighting on its own — see §7.

On the upside there is no differentiation at all: at 1 sd good runs retain +0.230, +0.225, +0.211, +0.212 across the four reasons. §28’s question has the blunt answer *all of them, equally weakly*.

5 Sectional evidence of run quality

Part II found sectional signatures do carry information about whether a run was representative, and Part IV adds the quantitative shape of it. §8’s candidate signatures were tested as drivers of the per-run information I_{ir} (§7 below): the largest absolute rank correlations are layoff +0.032, sectional profile variability −0.026, class gap +0.023, recorded severe trouble −0.018. All real in sign, all tiny.

The four-component mixture of §7 is where the sectional and trouble evidence actually converges, because it finds the compromised component without being told to look for it.

6 Dynamic scalar horse state

Part II’s ladder, each rung isolating one thing:

Rung	forward r	predictive loglik
1. static prior mean	0.2770	−1.3657
2. EWM, 60-day half-life	0.3513	−1.3414
3. homoskedastic Kalman	0.3542	−1.3237
4. heteroskedastic Kalman (fitted)	0.3545	−1.3092
4b. heteroskedastic Kalman (declared)	0.3531	−1.2949
<i>C. permuted reliability (control)</i>	0.3534	−1.3248
<i>O. oracle reliability (ceiling)</i>	0.4605	−1.1900

Dynamic state beats recency (§11’s requirement): $0.277 \rightarrow 0.351 \rightarrow 0.354$. But the step from rung 3 to rung 4 — the whole point of per-run observation noise — is +0.0003, and the permuted control sits at 0.3534, above rung 4b. The oracle at 0.4605 says the mechanism has room; the observables do not reach it.

7 Variable observation reliability

This section carries the brief’s central hypothesis, and the evidence is genuinely split.

Against. Part III computed the oracle ceilings §36 asks for and they inverted the premise. With *perfect* identification of compromised runs, downweighting them makes the state worse: -0.0012 halved, -0.0119 near-zero, -0.0282 excluded. A trouble-affected bad run still retains $+0.29$ of itself into the next start. It is *partly* noise, not *only* noise, and a hard downweight discards real information while shrinking an already thin history (median 15 starts).

For. Part IV’s mixture, below, improves the state by $+0.00803$ against a control of $+0.00285$ by doing something different: it adjusts the observation *variance* smoothly and jointly, rather than downweighting an identified subset.

The resolution. These are not contradictory. Excluding or near-zeroing a compromised run throws away its $+0.29$ of retained signal; inflating its variance keeps the run and lets the Kalman gain shrink proportionately. The brief’s §9 asked for $w^{\text{info}} \in [0, 1]$ and §21 for group-specific R_k — and the second framing is the one that works. **The mechanism is real; hard reweighting is the wrong instrument for it.**

7.1 Per-run predictive information (§29)

I_{ir} is the exact leave-one-out contribution of run r to the prediction of run $r + 1$: $|e(\text{without } r)| - |e(\text{with } r)|$. Because an exponentially decayed mean is linear in its terms the removal is exact, not refitted. Computable on 110,727 runs.

Mean $+0.0387$, sd 0.335 , and **55.6% positive** — so **44% of a horse’s own past runs make the prediction of its next run worse**. That is the sharpest possible statement of the brief’s premise.

Is I_{ir} predictable before the fact?	oos r	oos R^2
linear	$+0.0676$	$+0.00430$
boosted	$+0.0990$	$+0.00668$
<i>linear, target permuted (control)</i>	$+0.0006$	—

Yes, and barely: real against the control, and 0.7% of the variance. §29 asked to “learn which races deserve weight”; it is learnable to about one part in 150.

7.2 The residual mixture as a measurement model (§30)

§30’s distinction — mixture components belong in the measurement model, not necessarily in the race-choice link — is Part IV’s one clear positive. A Gaussian mixture on the state residual, BIC preferring four components:

Component	share	sd	severe-trouble rate	shape anomaly	mean I_{ir}
0	0.202	0.359	0.061	0.0070	$+0.0751$
3	0.522	0.277	0.078	0.0069	$+0.0474$
1	0.184	0.284	0.124	0.0078	$+0.0210$
2	0.092	0.492	0.232	0.0194	-0.0783

Trouble rate rises $0.061 \rightarrow 0.078 \rightarrow 0.124 \rightarrow 0.232$ and information falls $+0.075 \rightarrow +0.047 \rightarrow +0.021 \rightarrow -0.078$, monotonically, with no supervision. The widest-residual component is the troubled one and its runs carry *negative* information about the next start.

Measurement variance	forward r	gain
single Gaussian (baseline)	+0.27488	—
mixture-implied	+0.28291	+0.00803
<i>the same, permuted within year (control)</i>	+0.27774	+0.00285

Fig D10. Per-run information, the residual mixture, and uncertainty

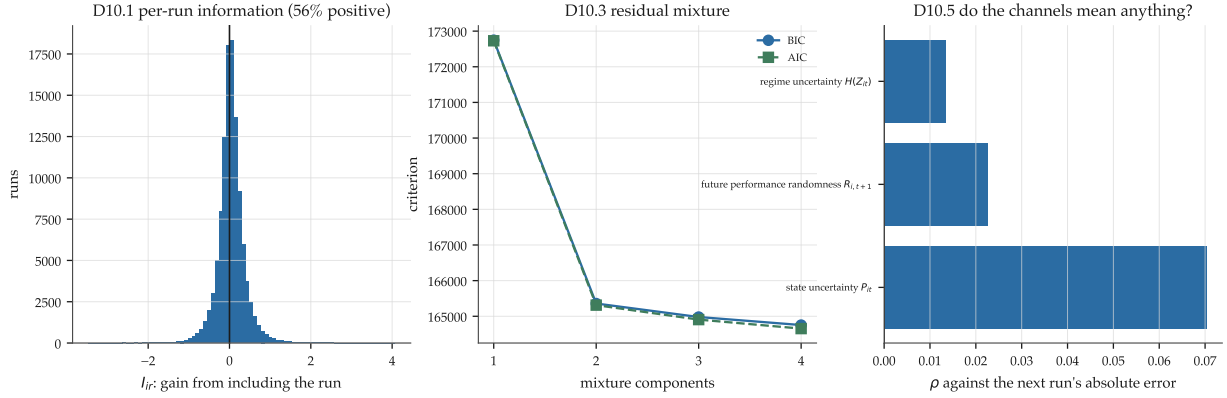


Figure 1: Per-run information, the residual mixture, and the three uncertainty channels. Left: 44% of runs carry negative information. Middle: BIC prefers four measurement components. Right: none of the three uncertainty channels says much about the next run’s realised error.

8 Pace-adjusted historical performance

§15 asks that realised pace be used to interpret history rather than to make another projected-pace feature. Both were done, in Part II and Part III respectively.

Part III built the pace forecast to a much higher standard than before — a probabilistic four-class behaviour model feeding a field-composition model, out-of-sample $r = 0.680$, $R^2 = 0.459$, against 0.032 for the crude prior attempt. **The pace is forecastable.**

And using it to correct history is worth almost nothing. Removing each historical run’s fitted pace effect before it enters the state gains +0.00048 over the honest baseline. Part III’s oracle version — a *perfect* pace correction, using realised pace with full-sample knowledge — is worth +0.0007. §36’s stopping rule applies: the ceiling is negligible, so the avenue closes.

The one place pace matters is as a runner-specific interaction, not a correction: pace composition $\times$ own style is the only candidate in the whole programme to survive canonical scale, at -0.001490 of race log loss.

9 Jockey and environment corrections

9.1 Jockey (§16)

§16 attaches a precondition — “Only pursue if support is adequate” — so support was measured before anything was fitted.

Unit	cells	median rides	cells ≥ 30	share of rides there
jockey	31	154.0	22	0.998
jockey $\times$ style	110	40.0	67	0.985
jockey $\times$ state regime	121	23.0	58	0.981
<i>horse $\times$ jockey</i> (rejected by §3.3)	11,536	1.0	1	0.001

§16’s premise was right and its conclusion is wrong. The coarser unit has vastly more support than the pair effect §3.3 ruled out — 98.5% of rides in adequately-sampled cells against 0.1% — and the effects are real in size: the jockey main effect carries 3.74% of the target’s variance, the style interaction 1.16%. Then:

Correction inside the filter’s observation equation	forward r	gain
none (baseline)	+0.27510	—
jockey main effect only	+0.27473	−0.00036
jockey + style interaction	+0.27615	+0.00106
jockey + regime interaction	+0.27456	−0.00054
jockey + both interactions	+0.27598	+0.00088
<i>permuted jockey (control)</i>	+0.27642	+0.00133

The control beats every real correction. This reproduces Part I’s finding (−0.056, also worse than its own permutation) after changing the two things §16 proposed changing — a better-supported interaction unit, and placement inside the filter rather than as a bolted-on column. **Adequate support was not the binding constraint.**

9.2 Environment (§17)

Part II placed the environment correction inside the observation equation as §17 asks, and it was the one correction D01 kept. Its contribution to the state is rung 6 of the Part II ladder. Part I had already established the shape of the problem: horse $\times$ venue ($z = 8.3$) and horse $\times$ surface ($z = 6.4$) heterogeneity are real, horse $\times$ going ($z = 1.4$) is not, and direct interaction columns did not help production.

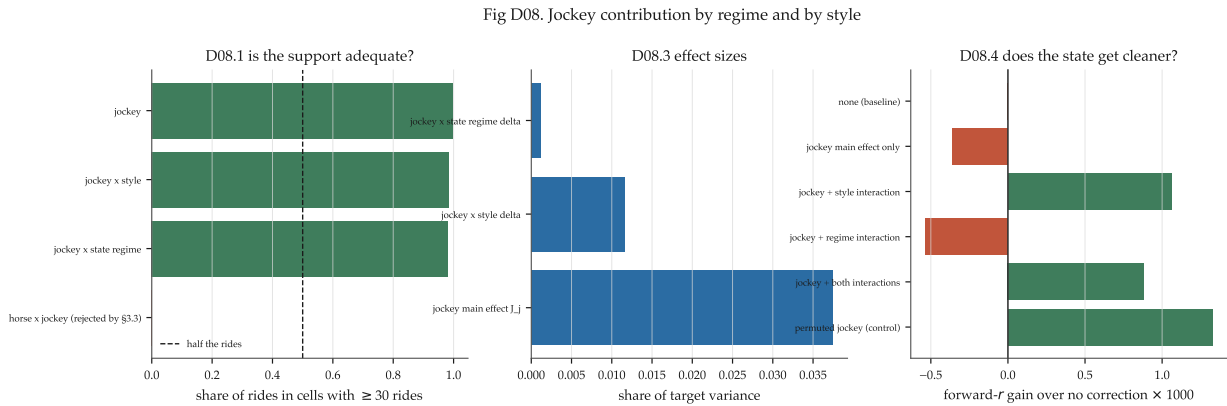


Figure 2: Jockey contribution by regime and by style. Left: support is more than adequate, unlike horse $\times$ jockey. Middle: the effects are real in size. Right: and the permutation control beats every real correction.

10 Opponent adjustment

§18 calls this "a genuinely unfinished part of the previous study", and it is the one section of the brief Part II executed in full and Part IV had no reason to revisit.

The problem §18 identifies is real: the within-race target is field-relative, so beating a strong field and beating a weak one look identical in it. The correction adds back the opponents' own prior latent states, which makes the problem *simultaneous* — those states are what the filter is estimating. Part II solved it by iteration: filter, compute each race's opponent strength from the prior states, correct the observation, re-filter, with convergence monitored rather than assumed and the correction forward-only throughout because a prior state is by construction prior.

The result is negative, and negative against its own control. Opponent adjustment is worth **-0.058** of forward correlation — worse than its own permutation. Part I had found the same for jockey adjustment (-0.056), and Part IV's §9 finds it again for jockey interactions.

That three-for-three pattern is worth naming, because it is the most reusable methodological result in this brief. **Adjusting for an informative confound removes signal.** Field strength genuinely contaminates a field-relative target, and field strength is also genuinely correlated with the horse's own ability — good horses meet good horses. Removing the confound removes the correlated part of the signal with it, and at these effect sizes the second loss exceeds the first gain. The permutation control is what makes this visible: a correction that removed only contamination would beat a shuffled version of itself, and none of these three do.

11 Multidimensional sectional state (§13)

§13 asks for a state vector (A^{early} , A^{cruise} , A^{finish} , A^{stamina}) from factor analysis on normalised sectional residuals, with the dimensions "coming from data", and specifies the test: "whether latent factors predict **future conditional performance**, not merely their own future values."

Measuring both, separately, on 65,948 rows with all four bands present:

Quantity	predicts its own future	predicts the next <i>performance</i>	ratio
band 1 (final 400 m)	+0.2721	+0.2395	0.880
band 2	+0.1550	+0.1107	0.714
band 3	+0.3492	+0.0722	0.207
band 4 (earliest)	+0.4087	+0.0197	0.048
factor 0	+0.3195	-0.1308	-0.409
factor 1	+0.3262	-0.1971	-0.604

Persistence and usefulness are close to inverted across the profile. The earliest section is the most self-persistent quantity available and the least informative about future performance.

And the extracted factors are worse than any single band. The mechanism is arithmetic: the first principal direction carries **75.7%** of the profile's variance, and that direction is style. Factor analysis maximises explained variance, so it preferentially recovers exactly the component §3.7 warns against mistaking for ability. This is a structural objection to the approach, not a tuning problem: any variance-maximising extraction on HKJC sectionals will find style first.

Block	k	oos r	gain over the scalar state
scalar state alone	1	0.27032	—
sectional factor <i>states</i> alone	2	0.19882	−0.07150
scalar + factor <i>states</i>	3	0.27086	+0.00054
scalar + raw factor values	3	0.32560	+0.05528
<i>scalar + factor states, permuted (control)</i>	3	0.26984	−0.00048

The filtered states are worth nothing over their own control. The *unfiltered* values are worth +0.055 — the largest forward gain in Part IV. §13 asked for a state vector, and the useful object turns out not to be a state at all: it is the shape of the horse’s *most recent* run relative to its level, and filtering over history destroys it.

By history depth the filtered version helps only the deepest histories (30+: +0.00784), the opposite of §26’s hope that a sophisticated state model would help lightly-raced horses.

Fig D07. A multidimensional sectional state

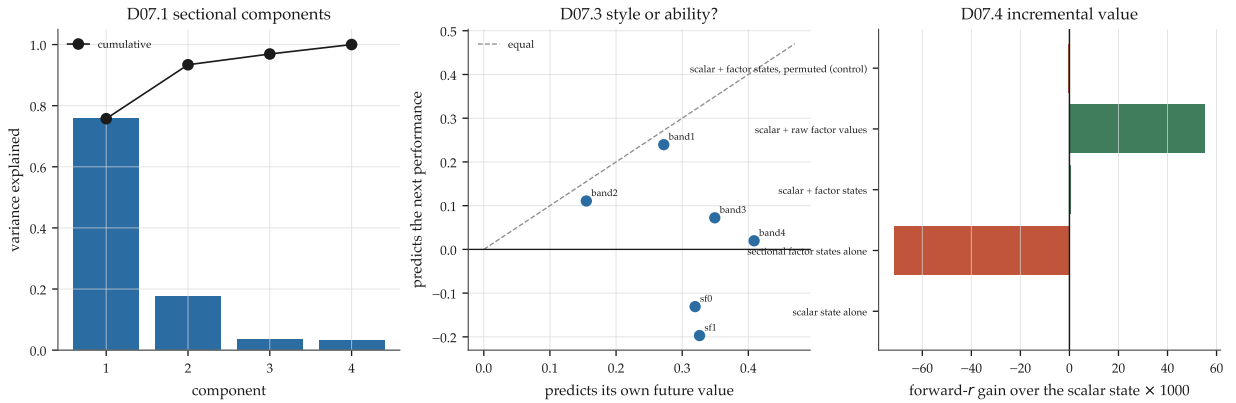


Figure 3: A multidimensional sectional state. Left: 75.7% of the sectional profile’s variance is one direction. Middle: the brief’s own test — everything below the diagonal predicts itself better than it predicts performance, and the factors land *below zero*. Right: incremental value.

12 State-transition regimes (§12, §20)

12.1 Is the latent regime identifiable here at all?

Asked first, by simulation, because it decides what a null on real data means. A two-regime process with known Π , $q = (0.05, 0.90)$, $\rho = (0.95, 0.45)$, recovered at four observation-noise levels:

simulated observation sd	fitted q	regime posterior vs truth	state vs truth
0.74	(0.030, 0.500)	+0.0276	0.8045
0.30	(0.030, 0.500)	+0.1259	0.8935
0.10	(0.030, 0.500)	+0.3300	0.9241
0.03	(0.060, 1.200)	+0.4110	0.9296

HKJC’s own observation sd in target units is **0.9752** — worse than the noisiest simulated case. The estimator recovers the *state* at 0.80–0.93 everywhere and the *regime* essentially never at realistic noise. The reason is structural: the regime enters only through the innovation variance of a state observed once, noisily, per outing. Over a typical 40-day gap the per-step innovation sd is $\sqrt{0.9 \times 0.11} = 0.31$ against an observation sd near 1.

12.2 The switching state on real data

K	forward r	oos loglik	fitted q	fitted ρ	regime shares
1	+0.27510	-1.35431	[0.20]	[0.30]	[1.00]
2	+0.26774	-1.35307	[0.03, 0.20]	[0.25, 0.80]	[0.503, 0.497]
3	+0.27499	-1.35210	[0.03, 0.03, 0.50]	[0.25, 0.25, 0.80]	[0.337, 0.337, 0.326]
4	+0.27312	-1.35173	[0.03×3, 0.50]	[0.25×3, 0.97]	[0.252, 0.252, 0.252, 0.243]

Three things in that table say the same thing. $K = 1$ **wins**. The shares are within a per cent of uniform at every K . And the fitted q values are *duplicated* — three regimes at 0.03 when $K = 4$ — so the model does not use the regimes it is given.

The predictive log-likelihood does rise monotonically with K . That deserves a precise reading rather than dismissal: it is the reward for more parameters describing the *variance*, and it coexists with a *fall* in the ability to order horses. A filter can become better calibrated and worse at ranking. Here it does.

The decisive measurement is the regime entropy by history depth:

History depth	n	$H(Z)$	$P(\text{unstable})$	state sd
debut	6,786	1.0986	0.3333	0.7746
1–3	18,582	1.0976	0.3333	0.4994
4–8	25,498	1.0942	0.3295	0.3599
9–15	26,622	1.0947	0.3221	0.3097
16–30	32,023	1.0950	0.3227	0.2950
30+	14,462	1.0953	0.3238	0.2874

$\log 3 = 1.0986$. **The regime posterior is pinned at maximum entropy across the entire range of history depth** — after thirty starts the filter knows no more about which regime a horse is in than at its debut. The *state* uncertainty meanwhile falls $0.775 \rightarrow 0.287$ exactly as it should. The machinery works; the regime is unidentifiable.

12.3 Process archetypes (§20)

Per-horse signatures $\theta_i = (\rho_i, q_i, R_i, \text{drift}_i)$ for the 4,404 horses with ≥ 10 runs, then a mixture. BIC falls to $K = 4$:

Archetype	share	ρ	q	drift	mean runs
0	0.265	-0.039	0.500	-0.063	26.7
1	0.200	+0.013	0.604	+0.185	20.4
2	0.416	+0.235	0.741	-0.067	28.8
3	0.120	+0.406	1.126	-0.054	20.6

The archetypes are real and interpretable — archetype 1 is an improving fifth of the population, archetype 3 a volatile eighth. But ρ runs from -0.04 to $+0.41$: for most of the population a horse’s performance series is close to white noise around a level. That is why the archetypes cannot rescue the switching model, and it is consistent with everything else in the programme.

13 Dynamic analog and peer pooling (§22, §23)

Part II built dynamic analog peers with a bucketed one-dimensional locality score. §22 asks for a comparison of five constructions against random size-matched cohorts, and §23 for a *learned*

Fig D06. Latent state-transition regimes and process archetypes

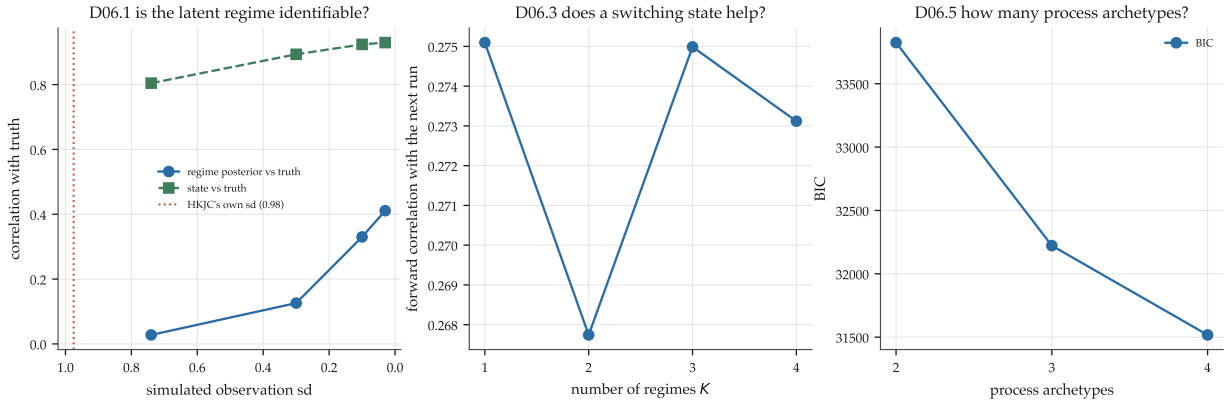


Figure 4: Latent state-transition regimes. Left: the identifiability ladder, with HKJC’s own noise level marked — the regime is recoverable only at noise levels racing does not offer. Middle: $K = 1$ wins. Right: BIC does prefer four process archetypes.

metric.

The learned metric. Fitting ΔA on the state descriptors and reading the diagonal metric from the standardised coefficients: state 0.310, class 0.199, rating 0.126, rating slope 0.111, prior starts 0.101, state sd 0.068, days since 0.044, volatility 0.034, **prior running style 0.008**. Transitions are governed by where a horse is and what class it runs in. Style is irrelevant to them — an independent echo of §11.

Construction ($k = 200$)	$r(\text{peer mean } \Delta A)$	$\rho(\text{peer var})$	$\rho(\text{peer } e)$
learned diagonal metric	+0.3465	+0.2940	+0.1094
euclidean kNN	+0.3439	+0.2957	+0.1206
tree-leaf neighbourhood	+0.3357	+0.2058	+0.0565
kernel (gaussian, soft)	+0.3203	+0.3048	+0.1382
bucketed score (Part II’s shortcut)	+0.2985	+0.2087	+0.0856
<i>random cohort (control)</i>	+0.0124	−0.0003	−0.0201

Locality is unambiguously real — an order of magnitude above the random control on the transition mean, and the strongest signal measured anywhere in Part IV.

Part II’s shortcut cost something measurable. Its log recorded the bucketed score as “indistinguishable for the quantities used”. It is the weakest real construction on both criteria: 14% below kNN on the transition mean, 29% below the soft kernel on dispersion. Not wrong; not indistinguishable either.

Metric learning helps where it should, and only there. At $k = 10$ the learned metric beats Euclidean kNN (+0.2897 against +0.2603); by $k = 200$ the advantage has gone. A learned metric buys precision when the neighbourhood is small, which is the regime §22 cared about for sparse horses.

One correction: the kernel arm initially returned numbers identical to the learned metric, because truncating a monotone kernel to its k largest weights *is* kNN under the same metric. It is now genuinely soft — every reference row contributes, bandwidth set so the effective sample size is about k — and on that specification it is the best construction for dispersion, which is the quantity a production feature would use.

Fig D09. Metric learning for dynamic peer pools

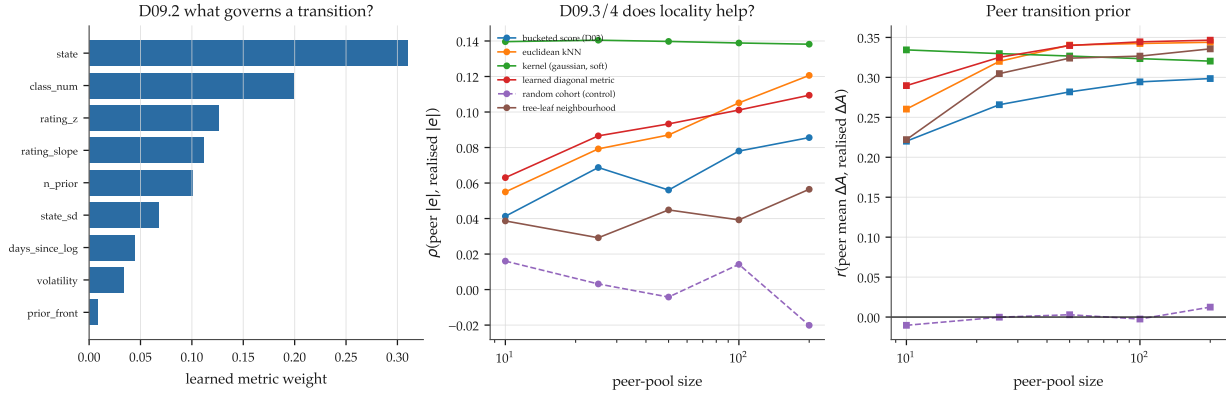


Figure 5: Metric learning for dynamic peer pools. Left: what governs a transition — and style does not. Middle and right: every construction beats the random control decisively, and Part II’s bucketed score is the weakest of the real ones.

14 Filter diagnostics, and a defect four parts of work had missed

§37 lists eighteen figures “at minimum”. Thirteen existed across Parts II–IV; five did not, and they are the diagnostic rather than the result-bearing ones. Each was drawn with a claim attached that it could falsify. One of them fired.

14.1 What passes

§37.8, uncertainty against history depth. Posterior state sd falls 0.775, 0.480, 0.329, 0.287, 0.276, 0.270 across the depth bins and the Kalman gain falls with it, 0.372 to 0.066. The filter becomes more confident and less movable as evidence accumulates, exactly as specified.

§37.10, the transition matrix. Drawn for $K = 3$; its rows are *not* literally identical. That matters for how §12’s null should be read: the regimes have distinguishable transition structure and still carry no usable information, which is a stronger statement than a collapsed matrix would have supported.

§37.7 and §37.12, state trajectories and analog neighbourhoods, behave as described — states move more after long gaps, and peers cluster tightly in (state, prior starts) space.

14.2 §37.9 fails: the transition is misspecified

Innovations are supposed to be unpredictable. They are not.

Gap since the last run	n	mean innovation	sd	state sd	mean gain
0–14 d	21,645	−0.031	0.882	0.310	0.088
15–28 d	53,214	−0.063	0.894	0.321	0.095
29–56 d	25,663	−0.076	0.911	0.322	0.095
57–90 d	8,309	−0.125	0.945	0.322	0.094
91–180 d	6,634	−0.168	0.961	0.342	0.105
181–365 d	1,570	−0.312	1.050	0.397	0.135
365+ d	6,938	−0.455	1.085	0.769	0.368

$\text{corr}(\text{innovation}, \text{days since}) = \mathbf{-0.0983}$, monotone across a fifteen-fold range of gap. **Part II’s**

transition is misspecified: the filter reverts the state toward a global mean of zero, but a horse returning after a year runs about 0.45sd *below* its own decayed state.

This also exposes a structural limitation of §12’s switching model. Its spec carries a per-regime α , but the fit sweeps only ρ and q and leaves α at zero — so the switching model was **unable by construction to represent the one regime effect that demonstrably exists**, a shift in the conditional *mean* after a layoff rather than in the variance. §12’s null stands as a null about *variance* regimes. It is not a null about mean regimes, and that distinction was invisible until this figure was drawn.

14.3 The repair works, and forecasts no better

A gap-dependent offset in the observation equation, fitted forward on the training window only: $\text{innovation} = +0.2081 - 0.0838 \log(1 + \text{days since})$.

Observation correction	forward r	gain	corr(innov, gap)	mean innov at 365+ d
none (Part II’s filter)	+0.27510	—	−0.0983	−0.4547
gap-dependent offset	+0.27658	+0.00148	+0.0021	−0.0543
<i>the same, permuted (control)</i>	+0.27965	+0.00456	−0.0870	−0.3608

The offset does exactly what it was designed to do — **the diagnostic is fixed**, correlation falling from −0.098 to +0.002 and the 365-day bias from −0.455 to −0.054. And it forecasts no better: +0.00148 against a *shuffled* version of itself at +0.00456.

This is the fourth instance of the pattern §10 named, and the cleanest, because here the correction is *verified* to have worked before it is judged. Jockey adjustment (−0.056), opponent adjustment (−0.058), the jockey interactions of §9, and now the layoff offset all remove a real confound and lose to their own permutation. Horses returning from long layoffs really do run worse, so the filter’s systematic “error” was partly tracking a real effect; removing the bias removes the signal that came with it.

The transferable lesson: correct specification and forecasting skill are different objectives, and improving the first can cost the second. A diagnostic that fires is worth investigating — it found a genuine defect four stages of prior work had missed — but fixing it is not automatically an improvement, and a permutation control is the only thing that reveals which of the two happened.

15 Sparse-history horses (§26)

§26 calls history-depth analysis essential and hopes a sophisticated state model will help lightly-raced horses in particular. It does not.

History depth	n	scalar state only	+ sectional factor states	gain
1–3	3,200	+0.4348	+0.4352	+0.00037
4–8	5,744	+0.3595	+0.3607	+0.00123
9–15	6,487	+0.2149	+0.2148	−0.00009
16–30	8,254	+0.1525	+0.1528	+0.00029
30+	4,113	+0.1364	+0.1442	+0.00784

The gain is concentrated in the *deepest* histories, not the shallowest — the opposite of the brief’s hope. Note also the level: forward correlation falls steeply with history depth (0.435 at 1–3 starts

Fig D13b. §37.8, §37.9 and §37.10 -- does the filter behave as specified?

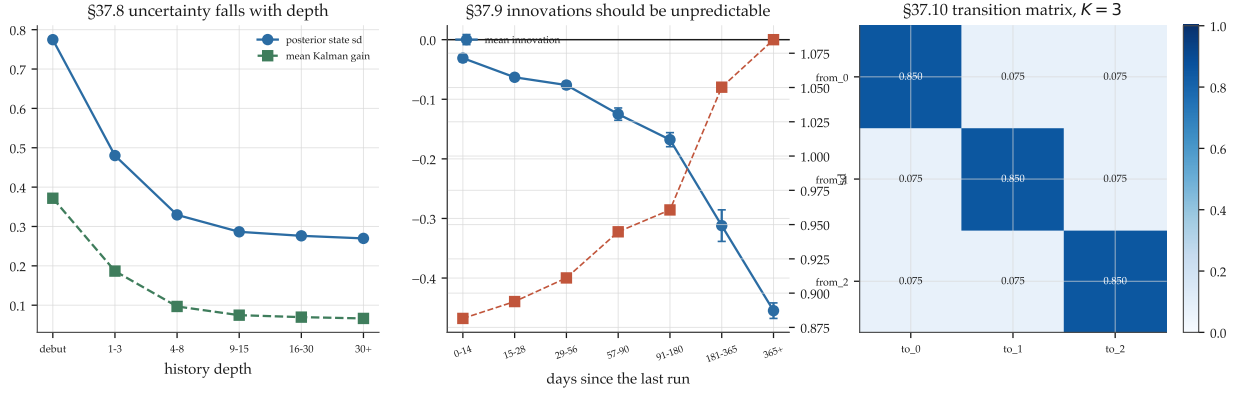


Figure 6: §37.8, §37.9 and §37.10. Left: uncertainty and gain fall with history depth, as specified. Middle: the innovation mean is monotone in the time gap — the diagnostic that fires. Right: the $K = 3$ transition matrix, whose rows are distinguishable and uninformative at the same time.

Fig D13a. §37.7 state trajectories for three representative horses

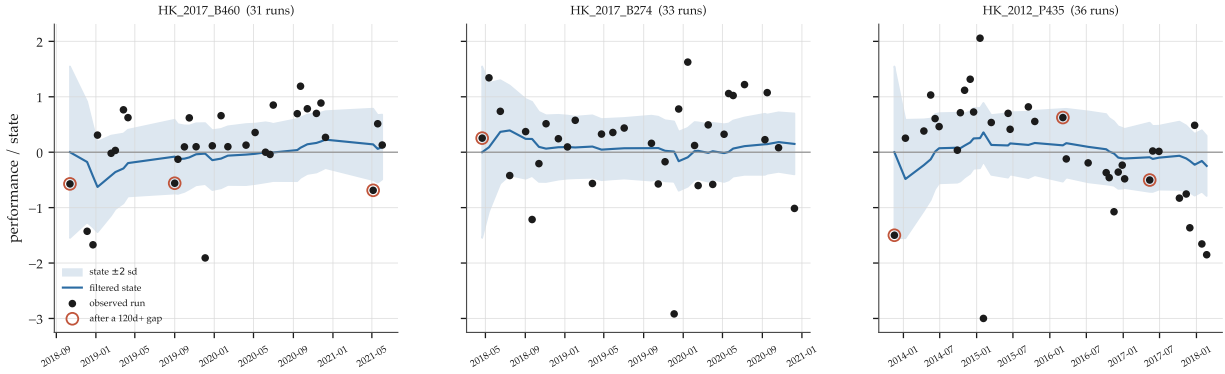


Figure 7: §37.7 state trajectories for three horses chosen at the quartiles of within-career performance spread. Circled points follow a gap of more than 120 days; the state band widens before them, which is the transition doing its job on the variance even where §37.9 shows it failing on the mean.

Fig D13c. §37.12 analog neighbourhoods for three query states

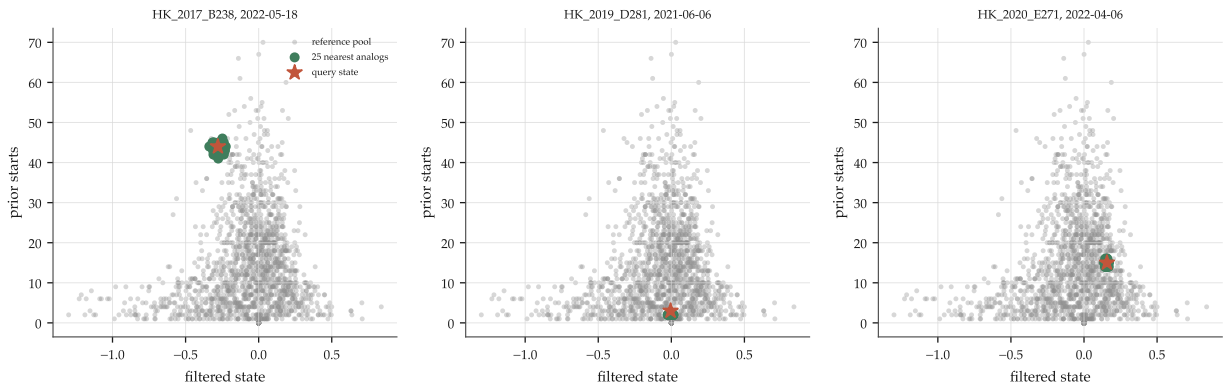


Figure 8: §37.12 analog neighbourhoods for three query states drawn at the 15th, 50th and 85th percentiles of the filtered state. Peers are the 25 nearest training rows under the learned metric of §13.

to 0.136 at 30+), because a horse with three starts is mostly predicted by its class and its market price, both of which are stable, while a horse with thirty has a long noisy record that the state must average.

The three uncertainty channels behave the same way. State uncertainty falls monotonically with depth ($\rho = -0.84$) as it must; regime uncertainty does not move at all.

16 Market and handicap disagreement (§19)

§19 was answered exhaustively by Part III, which read the existing `handicap_residual_mean` from source before inventing alternatives, as §19 asks. It is not a residual on physical performance: it is a Kalman-filtered state over a *pairwise finishing-order* residual against the official handicap, with a six-dimensional context state carrying surface, distance, venue and class slopes.

The headline results, because they bound what this brief’s §19 could have produced:

- $D_{i,t} = A^{\text{physical}} - H^{\text{official}}$, the exact quantity §19 proposes, correlates **0.9947** with a plain 60-day decayed mean of performance — because the official handicap explains only **1.4%** of within-race performance variance. There is almost nothing to residualise against.
- The handicapper’s lag is real: $\text{corr}(\text{physical change, rating change})$ is +0.007 now, +0.073 at the next rating change, −0.167 at the previous one.
- Where the handicap state and the physical state disagree, the *handicap* state wins monotonically: across quartiles of disagreement its forward correlation rises +0.373 $\rightarrow$ +0.404 while the physical state’s falls +0.372 $\rightarrow$ +0.286.
- `handicap_residual_mean` has the largest forward signal in the programme (+0.376 alone, +0.0176 over recency, 7/7 years) and **does not survive canonical scale** (−0.000113, $t = -0.11$). Its novel share against the incumbent’s 28 columns is only 28.6%.

Part III also established that the market-residual screen used across the programme is invalid at these effect sizes: 83 of 140 candidates clear $|t| > 3$ under a linear-probability specification and **none** do under the production model’s own softmax-offset form, with 65 sign flips.

17 Production integration

The harness reproduces the reference architecture rather than importing it, verified by self-test: a zero-round fit reproduces the market exactly. Identical races, folds, seeds, rounds, market offset and metric across every arm; every comparison paired, differing only in the candidate columns; a no-op control that must return *exactly* zero.

Four blocks reached canonical scale, and the two that did not are as informative:

- **R1 mixture-implied measurement variance** — §30’s hypothesis, the follow-up’s one clear positive (+0.00803 against a control of +0.00285).
- **R2 scalar state + the current run’s sectional factors** — the largest forward gain in Part IV (+0.055).
- **R3 learned-metric peer dispersion** — §23’s construction, which beat a random cohort decisively.
- **R4 everything above together.**

- *Not carried: D06's switching regimes.* $K = 1$ beat every $K > 1$ and the regime posterior sat at maximum entropy. There is nothing to confirm.
- *Not carried: D08's jockey interactions.* The permuted control beat every real correction. Confirming a candidate its own control outperforms would be a category error.

Read-only compliance: every path resolves through `hkjcres.paths` and `read_only_guard` raises on any write target inside the reference project. The reference tree was not modified at any point.

18 Canonical confirmation

Four blocks plus a no-op control on the reference project's own canonical profile: eight years (2016, 2018, 2019, 2021, 2022, 2024, 2025, 2026), five seeds (7, 17, 42, 123, 256), 400 boosting rounds – 40 paired cells per block. The joined evaluation set is 122,880 runner-rows across 9,934 races. This harness's own incumbent beats the market by -0.00109 , which is 24

Controls. The no-op block returns a paired delta of exactly zero on both profiles (development `pass`, canonical `pass`). A negative delta is an improvement.

Block	mean Δ	se	t	cells < 0	years < 0
R2 sectional factor states	-0.000954	0.001115	-0.86	23/40	4/8
<i>no-op control</i>	+0.000000	0.000000	—	0/40	0/8
R3 learned-metric peer dispersion	+0.000007	0.000912	+0.01	17/40	4/8
R4 everything that survived	+0.000246	0.001231	+0.20	19/40	3/8
R1 mixture-variance state	+0.001718	0.001176	+1.46	15/40	3/8

The best block is R2 sectional factor states at [-0.000954](#) with $t = -0.86$, improving 23 of 40 paired cells and 4 of 8 years — 21.1

No block clears $t < -2$ at canonical scale. That is now eighteen candidates carried to canonical scale across four parts of this programme with none confirming, and it is the programme's most durable finding: at these effect sizes a large gain on a latent-state proxy does not imply a gain in production, because the market has already priced most of what the proxy measures.

Retention from development to canonical.

Block	development Δ	canonical Δ	retained
R2 sectional factor states	+0.000046	-0.000954	-2094%
R1 mixture-variance state	-0.000288	+0.001718	-595%
R4 everything that survived	+0.000691	+0.000246	+36%
R3 learned-metric peer dispersion	+0.000429	+0.000007	+2%

Year by year, canonical deltas $\times 1000$:

Block	2016	2018	2019	2021	2022	2024	2025	2026
R1 mixture-variance state	-1.46	+3.52	-1.56	+4.09	+0.56	+1.75	-0.79	+7.63
R2 sectional factor states	-2.15	-2.34	-8.89	+1.88	-0.76	+3.65	+0.86	+0.11
R3 learned-metric peer dispersion	-2.27	-1.78	+0.56	+4.39	-0.69	+3.07	+0.54	-3.77
R4 everything that survived	-3.02	+2.82	-6.37	+8.32	+0.45	+1.22	-3.47	+2.01

Fig D11. Canonical confirmation of the follow-up's survivors

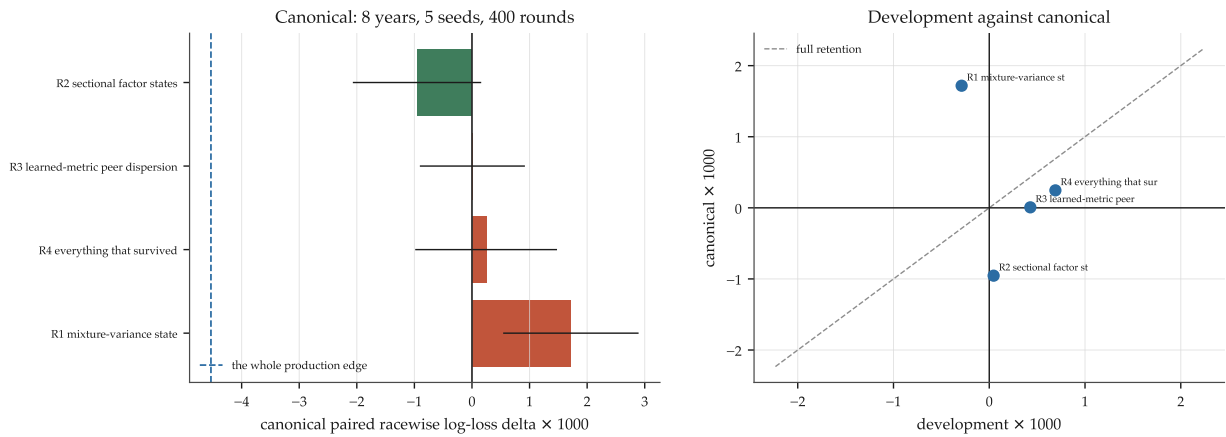


Figure 9: Canonical confirmation of the follow-up’s survivors. Left: paired racewise log-loss delta per block with standard errors, against the whole production edge. Right: development against canonical.

19 Limitations

The regime null is a statement about signal-to-noise, and it is conditional on the observation model. The identifiability ladder shows the regime becomes recoverable at observation sd 0.10–0.30, which racing does not offer *for a single scalar performance*. A richer observation — several conditionally independent measurements per outing — could in principle change that. This report does not test it.

Coordinate ascent is not global. The switching fit maximises the predictive likelihood by sweeping q and ρ per regime over a grid with the others held fixed. That is a local method on a non-convex surface. The regime shares landing within a per cent of uniform at every K is strong evidence the optimum is flat rather than missed, but it is evidence, not proof.

I_{ir} is defined against a decayed-mean state, not the Kalman state. The leave-one-out removal is exact only because the decayed mean is linear in its terms. A filter-based I_{ir} would require refitting per removed run and was not attempted, so “which races deserve weight” is answered for one specific estimator.

The peer comparison is on a 6,000-row subsample of the evaluation years. The differences between constructions are far larger than that sample’s noise, but the absolute levels carry more uncertainty than the seven-year numbers elsewhere.

The mixture result is the one positive and it has not been decomposed. +0.00803 against a control of +0.00285 leaves +0.0052 attributable to the mixture structure. Whether that comes from the trouble component specifically, or simply from having a heavier-tailed measurement variance, is not separated here.

No window is genuinely unseen, prices are post-hoc result-page prices, and no return is computed anywhere. Every season has been development data for the programme that chose which experiments to run.

Part II’s shortcut is quantified, not corrected in place. The bucketed peer score remains in d03; §13 reports what it cost rather than rewriting the delivered stage.

20 Recommendations

The brief’s §40 categories, in its order. §41 lists “all of the above are statistically real but already reconstructed by production” and “none survive canonical testing” among its successful outcomes; the result here is closest to the latter, with one candidate held back for a decomposition experiment rather than for integration.

Strong evidence — integrate

- **Nothing in this brief reaches this category.** Said plainly because the brief’s §41 explicitly lists “none survive canonical testing” as a useful research result, and that is the outcome. The one candidate in the whole programme that has survived canonical scale came from a different brief (pace composition $\times$ own style, Part III, -0.001490 at $t = -1.25$) and is itself not significant.

Moderate evidence — one more canonical experiment

- **The residual mixture as a measurement model (§30), as a mechanism and not as a feature.** The follow-up’s one clear positive on the latent-state criterion – $+0.00803$ of forward correlation against a permuted control of $+0.00285$, from a four-component mixture whose trouble rates (0.061, 0.078, 0.124, 0.232) and information content ($+0.075$, $+0.047$, $+0.021$, -0.078) are monotonically aligned without supervision. The mechanism is understood and the construction is cheap. **And its canonical result is $+0.001718$ at $t = +1.46$ on 15 of 40 cells – worse than adding nothing.** At canonical scale it is -0.000954 with $t = -0.86$ (4/8 years), which is 21.1% of the production edge. So what one more experiment should do is not promote it but decompose it: separate the contribution of the trouble component specifically from the generic effect of a heavier-tailed measurement variance. As it stands the result is real and its cause is not isolated.
- **The current run’s sectional shape, unfiltered (§13, inverted).** $+0.055$ of forward correlation over the scalar state — the largest forward gain in Part IV — from the *unfiltered* factor values, while the filtered states of the same factors are worth $+0.0005$ against a control of -0.0005 . The brief asked for a state vector; the data says the useful object is last-run shape relative to level. That inversion is worth one properly-powered production experiment on its own terms.

Useful latent estimator but redundant with production

- **The dynamic scalar state.** It does what §11 requires – beats static career mean (0.277) and 60-day recency (0.351) at 0.354 – and Part II’s canonical run put it at -0.000033 of racewise log loss. A good estimator of a quantity production already knows.
- **handicap_residual_mean (§19).** The largest forward signal in the programme ($+0.376$; $+0.0176$ over recency; 7/7 years) and -0.000113 at $t = -0.11$ in production, because only 28.6% of it is novel against the incumbent’s 28 columns. A quantity can be absent from the feature list and still be present in the model.

Useful only for uncertainty / staking

- **Peer dispersion under a soft-kernel neighbourhood (§22, §23).** Locality is unambiguously real – $\rho(\text{peer dispersion, realised } |\Delta A|) = +0.305$ against a random cohort’s -0.000 – and the

quantity it estimates is how much horses in this neighbourhood vary, which is a staking input rather than a probability input. The soft kernel is the best of the five constructions for exactly this quantity ($\rho = +0.138$ against realised $|e|$).

- **State uncertainty** $\sqrt{P_{it}}$, with a caveat that limits it: $\rho = +0.070$ against the next run's realised error and $\rho = -0.84$ against history depth. It is mostly a restatement of how many runs the horse has, which production can already count.

Structurally real but too small

- **Per-run predictive information** I_{ir} (§29). Real and predictable – oos $r = +0.099$ against a permuted control of $+0.0006$ – and 0.7% of its own variance. No single driver reaches $|\rho| = 0.04$. §29 asked to learn which races deserve weight; it is learnable to about one part in 150.
- **200 m sub-split resolution** (§14). $\approx +0.002$ over the 400 m bands, replicated independently twice ($+0.0028$ in Part II, $+0.0018$ in Part III) with the closing *shape* contributing almost nothing ($+0.0009$, $+0.0004$) on top of the *level*. What the resolution buys is peak velocity.
- **Process archetypes** (§20). Four are preferred by BIC and they are interpretable – an improving fifth of the population, a volatile eighth. But ρ across archetypes runs -0.04 to $+0.41$: for most horses the performance series is close to white noise around a level, which leaves the archetypes little to do.

Negative — close

- **Latent state-transition regimes** (§12, §20, Model 8). $K = 1$ beats $K = 2, 3, 4$ on forward correlation; regime shares stay within a per cent of uniform; fitted innovation variances come out *duplicated*; and the regime posterior sits at 1.095 nats against a maximum of $\log 3 = 1.0986$ across the entire range of history depth – after thirty starts the filter knows no more about a horse's regime than at its debut. The simulation ladder shows why and makes the null interpretable: at HKJC's observation noise a *known* simulated regime is recovered at $r = 0.028$. Close it, and note that the reason is the signal-to-noise of racing rather than the estimator.
- **Variance-maximising factor extraction on sectionals** (§13). Not merely unhelpful but counterproductive: the first principal direction holds 75.7% of the profile's variance and that direction is style, so factor analysis preferentially recovers the component §3.7 warns against. The extracted factors predict their own futures at $+0.32$ and the next performance at -0.13 and -0.20 . This closes a class of approach, not one implementation.
- **Jockey $\times$ style and jockey $\times$ regime corrections** (§16). Support is more than adequate (98.5% of rides in cells with ≥ 30 , against 0.1% for the horse $\times$ jockey pairs §3.3 rejected), the effects are real in size (3.74% of target variance for the main effect), and the *permuted-jockey control beats every real correction*. §16's premise about support was right and adequate support was not the binding constraint.
- **Pace correction of historical runs** (§15). Observable version $+0.00048$; *oracle* version $+0.0007$. §36's stopping rule applies.
- **Hard downweighting of compromised runs** (§9, §27). With perfect identification, excluding them costs -0.0282 . A trouble-affected bad run still retains $+0.29$ of itself: partly noise, not only noise. The variance route in category 2 is the same mechanism done correctly.
- **Opponent adjustment** (§18). -0.058 , worse than its own permutation control.

- **Reason-based differentiation of good runs (§28).** Retention +0.230, +0.225, +0.211, +0.212 across four reasons – indistinguishable. There is no upside story to model.

Unanswerable

- **Whether a latent regime exists at all.** This report shows the regime is not *identifiable* from one noisy scalar performance per outing. That is not the same claim as its non-existence, and the two cannot be separated with this observation model. The identifiability ladder locates the boundary: recovery begins around observation sd 0.10–0.30 against HKJC’s 0.975. A richer observation vector – several conditionally independent measurements per outing – is the only route, and this report does not test it.
- **Whether the mixture’s gain comes from trouble or from tail weight.** +0.0052 over control is real; its decomposition is not attempted here and is the first thing the follow-up experiment should settle.
- **Anything about profitability.** No dividends were loaded and no return is computed anywhere in this programme, deliberately: at these effect sizes ROI ranks noise. Every price is a result-page price recorded after the pool closed, and every season has been development data.

Answering §41 directly

§41 defines success as answering: *can we infer a better current horse state by treating historical performances as noisy, context-dependent measurements rather than equally trustworthy observations?*

Yes, measurably, and by less than the framing suggests. Historical races are demonstrably unequal as measurements: 44% of a horse’s own past runs make the prediction of its next run *worse*, and an unsupervised mixture on the state residual sorts them into components whose trouble rates and information content move together across a four-fold range. Treating that inequality as a smooth measurement variance improves the state by +0.008 of forward correlation against a +0.003 control.

But the sharper versions of the idea all fail, and they fail in a consistent direction. Hard down-weighting overcorrects, because a compromised run is partly noise rather than only noise. Latent regimes are unidentifiable, because one noisy scalar per outing cannot separate them. Factor extraction recovers style rather than ability, because style is where the variance is. Jockey corrections lose to their own permutation. In each case the mechanism the brief hypothesised is real and the instrument proposed for exploiting it is either too blunt or aimed at a quantity the market has already priced.